Li Zhang

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Education

University of Pittsburgh Pittsburgh, U.S.A.
Doctor of Philosophy in Intelligent Systems *Aug. 2024 - Apr. 2029*

- Research: AI & Law
- Advisor: Prof. Kevin Ashley
- GPA: 4.0/4.0
- Courses: Large Language Models, Natural Language Processing, Machine Learning

Tsinghua University Beijing, China
Master of Computational Law *Sep. 2019 - Jun. 2022*

- Research: Computational Law & Business Law
- Advisor: Prof. Shangshang Liang

Australian National University Canberra, Australia
Study Abroad Student in College of Engineering and Computer Science *Jul. 2017 - Dec. 2017*

- GPA: 6.5/7.0

China University of Mining and Technology Jiangsu, China
Bachelor of Electrical Engineering *Sep. 2015 - Jun. 2019*

- GPA: 90.91/100, Ranking: 2/278

Projects

LLM Alignment and Applications in Legal Reasoning Pittsburgh, U.S.A
***Li Zhang**, Morgan Gray, Kevin Ashely* *Aug. 2024 –*

- Created a multi-agent framework to generate reliable legal arguments through collaboration and reflection mechanisms, achieving a decent performance in hallucination accuracy and abstention rate.
- Constructed vector database, developed and deployed a multi-agent LLM framework and for legal reasoning and document retrieval. Led collaboration with legal experts to establish evaluation metrics and validate model outputs against expert-defined reasoning standards.

ML-Based Prediction of Academic Performance Pittsburgh, U.S.A
***Li Zhang**, Morgan Gray, Jessica Macaluso, Kevin Ashely, Scott Fraundorf* *Aug. 2024 –*

- Developed a machine learning model that achieved a decent accuracy for law school academic performance using feature engineering and Random Forest regression, enabling early identification of at-risk students for targeted interventions.
- Conducted comprehensive feature selection and statistical analysis that revealed skill-specific predictors of academic success, providing actionable insights for legal education pedagogy and personalized learning approaches.

Automated Legal Citation Correction through ML and NLP Beijing, China
Li Zhang *Jun. 2021 – June. 2022*

- Built a DynamoDB repository for structured storage of labeled citation errors, enabling iterative model refinement. Achieved 95% precision in detecting inconsistencies through feature engineering (e.g., regex patterns, contextual embeddings) and hyperparameter optimization.
- Designed and deployed a scalable citation-correction system to train and host the NLP model. Partnered with legal researchers to validate outputs against the Bluebook standard.

AI-driven Open-source Compliance and Strategy Beijing, China
***Li Zhang**, Shangshang Liang* *Jan. 2020 – May. 2020*

- Developed a machine learning-based risk assessment system to automate compliance checks for 20+ open-source licenses (e.g., GPL, Apache). Leveraged NLP techniques to parse license terms and flag high-risk clauses, reducing manual review time by 60% while maintaining high accuracy.
- Trained a classification model to categorize FOSS projects by compliance risk levels, using features like license type, dependencies, and jurisdiction. Collaborated with legal experts to label datasets and validate outputs.

Publications

- **Li Zhang**, Kevin D. Ashley: *"Mitigating Manipulation and Enhancing Persuasion: A Reflective Multi-Agent Approach for Legal Argument Generation", LCIC@ICAIL 2025*, 2025
- **Li Zhang**, Morgan Gray, Jaromir Savelka, Kevin D. Ashley: *"Measuring Faithfulness and Abstention: An Automated Pipeline for Evaluating LLM-Generated 3-ply Case-Based Legal Arguments", ASAIL@ICAIL 2025*, 2025
- Morgan Gray, **Li Zhang**, Kevin Ashely: *"Generating Case-Based Legal Arguments with LLMs", In Proceedings of the 4th ACM Computers and Law Symposium*, 2025
- Weinan Liu, Boyuan Peng, Xiaoyu Liu, Feifan Ren, **Li Zhang***(corresponding): *"Intelligent Proximate Analysis of Coal Based on Near-Infrared Spectroscopy", Journal of Applied Spectroscopy*, 2021
- Meng Lei, **Li Zhang***(corresponding), Ming Li, Huiyu Chen, Xuan Zhang: *"Near-infrared Spectrum of Coal Origin Identification Based on SVM Algorithm", In Proceedings of Chinese Control Conference*, 2018
- Meng Lei, Ning Bian, Ming Li, Lianshuai Sha, Haowen Fu, **Li Zhang** *"A kind of water area refuse automated cleaning ship and automatic cleaning method based on machine vision", Patent Publication No. CN106741683A*, 2017

Talks

- *"Mitigating Manipulation and Enhancing Persuasion: A Reflective Multi-Agent Approach for Legal Argument Generation", Talk at ResearchTrend.AI*, 2025
- *"Enhancing Legal Argument Generation: From Basic Schemes to Trustworthy Multi-Agent Systems", Talk at ISP Forum, the University of Pittsburgh*, 2025

Skills

- **Programming:** Python (Pandas, NumPy, PyTorch, TensorFlow and Scikit-learn), SQL, JavaScript
- **Cloud Technologies:** SageMaker, DynamoDB and API Gateway
- **Others:** LaTeX, Markdown, Linux, Shell and Anaconda

Selected Awards & Honors

- SCI Fellowship, University of Pittsburgh, 2024
- Dean's List (top 1%), Tsinghua Law School, 2022
- Law School Fellowship, Tsinghua Law School, 2021
- Dean's List (top 1%), China University of Mining and Technology, 2019
- Study Abroad Scholarship (1 out of 180), China University of Mining and Technology, 2017